

There are only three row operations we've defined in lecture

1) $\lambda R_i + R_j \rightarrow R_j$: Multiply one row by a scalar and add that row to another row, replacing that row.

Note: There is no such thing as division in a vector space, it was never defined. There is also no such thing as subtraction in a vector space, also never defined.

Ex// $(3)R_1 + R_2 \rightarrow R_2$ is a well-defined row operation

Non-Ex// $R_2 - 2R_1 \rightarrow R_1$ is not a ~~vector space~~ row operation.

Non-Ex// $R_2 - 2R_1 \rightarrow R_2$ is deceptive and inaccurate. There is no such thing as subtraction in a vector space.

2) $R_i \leftrightarrow R_j$: Swap the position of rows.

3) $\lambda R_j \rightarrow R_j$: Multiply a row by a scalar.

Ex// $5R_2 \rightarrow R_2$ is a well-defined row operation

Non-Ex// $\frac{R_2}{5} \rightarrow R_2$ is not well-defined. There is no such thing as division in a vector space.

When trying to solve a matrix system, or linear algebraic system of equations, it's most optimal to perform the above operations to put that matrix system into Reduced Row Echelon Form.

Let's do an example to illustrate how to correctly and efficiently apply these.

Solve the system of equations; write your answer in vector form.

$$\begin{aligned} V - W - X - 2Y - Z &= 2 \\ 3V - W + X - Y + 2Z &= 5 \\ 5V - 2W + X - 3Y + 3Z &= 10 \\ 2V - W - 2Y + Z &= 5 \end{aligned}$$

It would be much more clear if we turn this / put this into an Augmented Matrix Form.

$$\left(\begin{array}{ccccc|c} 1 & -1 & -1 & -2 & -1 & 2 \\ 3 & -1 & 1 & -1 & 2 & 5 \\ 5 & -2 & 1 & -3 & 3 & 10 \\ 2 & -1 & 0 & -2 & 1 & 5 \end{array} \right)$$

The columns correspond to the variables $V, W, X, Y,$ and Z . The far right column is the $\vec{b}$ vector in the equation $A\vec{x} = \vec{b}$

$$\text{Note } A\vec{x} = \begin{pmatrix} 1 & -1 & -1 & -2 & -1 \\ 3 & -1 & 1 & -1 & 2 \\ 5 & -2 & 1 & -3 & 3 \\ 2 & -1 & 0 & -2 & 1 \end{pmatrix} \begin{pmatrix} V \\ W \\ X \\ Y \\ Z \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \\ 10 \\ 5 \end{pmatrix} = \vec{b}$$

The objective is to find $\vec{x} = \begin{pmatrix} V \\ W \\ X \\ Y \\ Z \end{pmatrix}$

Identify the position where you want your first pivot; Usually it will be the top left element in the augmented matrix. Use ROW OPERATIONS TO ELIMINATE EVERY ELEMENT BELOW THAT PIVOT. (If that element is not 1, and there is a 1 somewhere below, use Row Operation 2 to put the 1 in that position, it will make life easier).

For this first stage, perform the following:

$$\begin{aligned} -3R_1 + R_2 &\rightarrow R_2 \\ -5R_1 + R_3 &\rightarrow R_3 \\ -2R_1 + R_4 &\rightarrow R_4 \end{aligned}$$

Cognitively, you're multiplying a row by an element so that if you ADD that row to another you will get zero for that element you wish to eliminate; this is usually the element below the pivot.

$$\left(\begin{array}{ccccc|c} 1 & -1 & -1 & -2 & -1 & 2 \\ 0 & 2 & 4 & 5 & 5 & -1 \\ 0 & 3 & 6 & 7 & 8 & 0 \\ 0 & 1 & 2 & 2 & 3 & 1 \end{array} \right)$$

↑
Note, stage 1 complete. Move to the next column underneath that row.

Note, the second column (ignoring the first row, why?) has a 1; swap rows to put that 1 in the next pivot's position.

$$R_2 \leftrightarrow R_4$$

$$\left(\begin{array}{ccccc|c} 1 & -1 & -1 & -2 & -1 & 2 \\ 0 & 1 & 2 & 2 & 3 & 1 \\ 0 & 3 & 6 & 7 & 8 & 0 \\ 0 & 2 & 4 & 5 & 5 & -1 \end{array} \right)$$

Strategy: That move was not necessary, it simply will allow us to compute more simply.

Choose row operations to eliminate the 3 and 2 below the next pivot.

$$-3R_2 + R_3 \rightarrow R_3$$

$$-2R_2 + R_4 \rightarrow R_4$$

$$\left(\begin{array}{ccccc|c} 1 & -1 & -1 & -2 & -1 & 2 \\ 0 & 1 & 2 & 2 & 3 & 1 \\ 0 & 0 & 0 & 1 & -1 & -3 \\ 0 & 0 & 0 & 1 & -1 & -3 \end{array} \right)$$

↑
Stage 2 complete. As it may sometimes happen, we got more completed.

Note the repeated row!

$$(-1)R_3 + R_4 \rightarrow R_4$$

Note! No subtraction, adding a negative!

Date

$$\left(\begin{array}{ccccc|c} 1 & -1 & -1 & -2 & -1 & 2 \\ 0 & 1 & 2 & 2 & 3 & 1 \\ 0 & 0 & 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

This matrix is in Row Echelon Form. From here we may quickly identify much information.

Copied over...

$$\left(\begin{array}{ccccc|c} \textcircled{1} & -1 & -1 & -2 & -1 & 2 \\ 0 & \textcircled{1} & 2 & 2 & 3 & 1 \\ 0 & 0 & 0 & \textcircled{1} & -1 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

The pivots are circled. Here is all the information we can glean:

$$\begin{aligned} \text{Rank}(A) &= 3 \\ \text{Rank}(A|\vec{b}) &= 3 \end{aligned}$$

} From the Kronecker - Capelli Theorem, the system is consistent. What does that mean?

The basic variables are V , W , and y since they correspond to the first, second, and fourth columns where the pivots are.

The free variables are x and z .

By the Rank-Nullity Theorem, the Nullity (what's that?) is 2, which means we can expect two linearly independent vectors as solutions to the homogeneous equation, $A\vec{x} = \vec{0}$.

As you see, a lot can be learned! But, to solve the system, we have more work. We need to put the matrix into RREF to solve.

Start from the right-most pivot and eliminate the ~~row~~ entries above ~~and~~ it.

$$-2R_3 + R_2 \rightarrow R_2$$

$$2R_3 + R_1 \rightarrow R_1$$

$$\begin{pmatrix} 1 & -1 & -1 & 0 & -3 & | & -4 \\ 0 & 1 & 2 & 0 & 5 & | & 7 \\ 0 & 0 & 0 & 1 & -1 & | & -3 \\ 0 & 0 & 0 & 0 & 0 & | & 0 \end{pmatrix}$$

Stage 1 of Gauss-Jordan "backwards" complete.

Continue w/ the next pivot from the right.

$$(1) R_2 + R_1 \rightarrow R_1$$

$$\begin{pmatrix} 1 & 0 & 1 & 0 & 2 & | & 3 \\ 0 & 1 & 2 & 0 & 5 & | & 7 \\ 0 & 0 & 0 & 1 & -1 & | & -3 \\ 0 & 0 & 0 & 0 & 0 & | & 0 \end{pmatrix}$$

This augmented matrix is in Reduced Row Echelon Form!

From the text... the definition of RREF

1) Zero rows, if any, are at the bottom. ✓

2) The leftmost ~~entry~~ non-zero entry of each of the non-zero rows equals 1. ✓

3) Each pivot is further to the right than the pivot in the row above it. (Stairs) ✓

4) Each pivot is the only non-zero entry in its column. ✓

The reason for this form is that the above are equations w/ ~~exactly~~ at most one basic variable each!

Now, put the above into equation form, leaving the augmented matrix.

$$\begin{aligned} V + X + 2Z &= 3 \\ W + 2X + 5Z &= 7 \\ Y - Z &= -3 \end{aligned}$$

One basic variable each, in terms of the free variables.

Remember, free variables are free, they can take on any value. Consider them to be parameters.

Solve for the basic variables in terms of the free variables.

$$V = 3 - x - 2z$$

$$W = 7 - 2x - 5z$$

$$y = -3 + z$$

Let's not lose sight of what we needed! We always needed to solve for the vector $\vec{x}$ (on p. 2 of this pdf). Thus, we should replace the basic variable w/ what they are above.

$$\vec{x} = \begin{pmatrix} V \\ W \\ x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3 - x - 2z \\ 7 - 2x - 5z \\ x \\ -3 + z \\ z \end{pmatrix}$$

← Note that x and z are free variables, so they stay put. Remember, they are parameters now, they can be anything in $\mathbb{R}$.

Split this vector into the parts w/ ~~free~~ free variables, and those without.

$$\vec{x} = \begin{pmatrix} 3 \\ 7 \\ 0 \\ -3 \\ 0 \end{pmatrix} + x \begin{pmatrix} -1 \\ -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + z \begin{pmatrix} -2 \\ -5 \\ 0 \\ 1 \\ 1 \end{pmatrix} \quad \text{Final solution}$$

Because we have a linear system, we know the solution must have the form $\vec{x} = \vec{x}_h + \vec{x}_p$.

Observe

$$\vec{x}_h = \text{homogeneous solution} = x \begin{pmatrix} -1 \\ -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + z \begin{pmatrix} -2 \\ -5 \\ 0 \\ 1 \\ 1 \end{pmatrix}$$

Two linearly independent ~~subspace~~ vectors that form a basis of the Null space.

$$\vec{x}_p = \text{particular solution} = \begin{pmatrix} 3 \\ 7 \\ 0 \\ -3 \\ 0 \end{pmatrix}$$

Strategy! You can confirm these are the solutions you claim by plugging them into the original equation $A\vec{x} = \vec{b}$

$$\begin{pmatrix} 1 & -1 & -1 & -2 & -1 \\ 3 & -1 & 1 & -1 & 2 \\ 5 & -2 & 1 & -3 & 3 \\ 2 & -1 & 0 & -2 & 1 \end{pmatrix} \begin{pmatrix} -1 \\ -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 + 2 - 1 \\ -3 + 2 + 1 \\ -5 + 4 + 1 \\ -2 + 2 + 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \checkmark$$

$$\begin{pmatrix} 1 & -1 & -1 & -2 & -1 \\ 3 & -1 & 1 & -1 & 2 \\ 5 & -2 & 1 & -3 & 3 \\ 2 & -1 & 0 & -2 & 1 \end{pmatrix} \begin{pmatrix} -2 \\ -5 \\ 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 + 5 - 2 - 1 \\ -6 + 5 - 1 + 2 \\ -10 + 10 - 3 + 3 \\ -4 + 5 - 2 + 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \checkmark$$

and

$$\begin{pmatrix} 1 & -1 & -1 & -2 & -1 \\ 3 & -1 & 1 & -1 & 2 \\ 5 & -2 & 1 & -3 & 3 \\ 2 & -1 & 0 & -2 & 1 \end{pmatrix} \begin{pmatrix} 3 \\ 7 \\ 0 \\ -3 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 - 7 + 6 \\ 9 - 7 + 3 \\ 15 - 14 + 9 \\ 6 - 7 + 6 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \\ 10 \\ 5 \end{pmatrix} \quad \checkmark$$

J.G. ✓